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Model Question, 2079

Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC461)	**Pass Marks: 24**
(Advanced Database)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. Explain enhanced entity relationship model in detail. What is aggregation? [8+2]
2. What is distributed database? Explain data fragmentation techniques in detail with suitable example. [2+8]
3. Why do we need query optimization in databases? Compare heuristic query optimization with cost-based query optimization. [3+7]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Explain specialization and generalization with example. [5]
5. What are the benefits of object-oriented database? What is object identity? [3+2]
6. Explain object query language (OQL) with suitable example. [5]
7. What is query tree? Why do we need this tree in query processing? [2+3]
8. Explain different techniques for distributed database design. [5]
9.. Explain document-based NOSQL system with example. [5]
10. Explain multimedia database in brief. What are the different applications of multimedia database? [2.5+2.5]
11. Why do we need temporal database? Explain different time dimensions in this database. [2+3]
12. Write short notes on: [2 x 2.5=5]
    a. Indexing
    b. MapReduce

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC463)	**Pass Marks: 24**
(Advanced Networking with IPv6)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. What do you mean by programmable networks? Compare the programmable IPv6 network approach with traditional IPv4 networking. [5+5]

2. How does different IPv6 addresses are structured? Explain with example. [10]
3. What is SLAAC? Discuss the ICMPv6 message exchange process for SLAAC. [3+7]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Discuss the IPv6 security policy database and security association database. [5]
5. How does RIP for IPv6 work? Discuss. [5]
6. Explain the recent techniques of IPv6 network migration. [5]
7. Explain the challenges and risks of SoDIP6 network deployment. [5]
8. Discuss about PMTU Discovery. [5]

9.. What are the features of software defined IPv6 networks? Explain. [5]

10. Discuss about RTE format with example. [5]
11. Discuss about PIM-SM for IPv6. [5]
12. Explain about Open vSwitch. [5]

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC464)	**Pass Marks: 24**
(Distributed Networking)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1.	Define distributed system. Explain the client server model in distributed computing system.	[2+8]
2.	Discuss client server interoperability. Describe about communication paradigm.	[2+8]
3.	Explain the different types of attack in distributed network.	[10]

Section B

Attempt any eight questions. (8 × 5 = 40)

4.	Explain the characteristics of distributed network system.	[5]
5.	What are the techniques to achieve reliability? Describe.	[5]
6.	How does message is passed over network?	[5]
7.	Describe the communication protocol.	[5]
8.	Discuss about fault tolerance in case of distributed network.	[5]
9.	What are the possible security issues? Explain.	[5]
10.	How do you design and implement distributed shared memory.	[5]
11.	Describe about grid and P2P computing.	[5]
12.	Explain about different replication techniques.	[5]

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC465)	**Pass Marks: 24**
(Game Technology)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. What are different roles of a game designer? Explain formal elements of a game in detail. [4+6]
2. Why prototyping lies at the heart of good game design? Explain different methods of prototyping in detail. [3+7]
3. Explain different stages of game development. What is agile project planning? [7+3]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Why communication and teamwork are most important skills in game development? [5]
5. Explain dramatic elements in game development. [5]
6. How the elements of games fit together to form playable systems? Explain. [5]
7. Why brainstorming is a powerful skill in game development? [5]
8. What are different prototyping tools used in game development? Explain. [5]

9.. What is play matrix? How this matrix is used in playtesting? [2+3]

10. What is the size of game industry? What are the different distribution platforms for games? [2.5+2.5]
11. How do you pitch your own original ideas to publishers? What is pitch process? [3+2]
12. Write short notes on: [2 x 2.5=5]
    a) Game tuning
    b) Concept art

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC466)	**Pass Marks: 24**
(Distributed and Object Oriented Database)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. Define distributed database. Explain different distributed database management system architectures. [2+8]
2. Explain different layers of distributed query processing. What s data localization? [8+2]
3. Explain distributed concurrency control. How do you manage deadlock in distributed database environment? [5+5]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Explain benefits of distributed databases. What is autonomy? [3+2]
5. What is data fragmentation? Compare horizontal fragmentation with vertical fragmentation using suitable example. [1+4]
6. What is distributed query processing? What are the objectives of this query processing? [2.5+2.5]
7. Explain lock-based concurrency control technique in distributed environment. [5]
8. Why do we need object-oriented database? What is object structure? [2.5+2.5]
9. What is complex object? Explain type hierarchies and inheritance. [2+3]
10. Explain object definition language with example. [5]
11. What are the design issues related with distributed databases? Explain. [5]
12. Write short notes on: [2 x 2.5=5]
    a) Allocation
    b) OQL

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC467)	**Pass Marks: 24**
(Introduction to Cloud Computing)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. Define cloud. Describe the evolution of cloud. Mention the advantages of using cloud computing. [10]
2. Describe the services provided under cloud computing. What are the benefits of virtualization? [6+4]
3. What is Map-Reduce Programming? Describe how enterprise batch processing is done using map-reduce? [4+6]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Describe cloud service requirements. [5]
5. Differentiate public cloud from private cloud. [5]
6. Discuss the different types of hypervisors. [5]
7. How thread is different from task? How thread programming is done? [3+2]
8. Discuss the cloud security issues. [5]

9.. What is bucket in Amazon simple storage service? How addressing of a bucket is done? [2+3]

10. Describe how cloud computing is used in business and consumer applications. [5]
11. How virtual machine security is enforced? [5]
12. Describe the types of services hosted in Aneka container. [5]

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Bachelor Level/ Fourth Year/ Eighth Semester/Science — **Full Marks: 60**
Computer Science and Information Technology (CSC468) — **Pass Marks: 24**
(Geographical Information System) — **Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. Define geographical object and field. Differentiate between raster and vector data models with illustrations. [10]
2. What are different coordinate and projection systems used in GIS mapping. Explain in brief. [10]
3. How Remote Sensing is different from GPS? Explain how Remote Sensing works? Differentiate between active and passive sensor? [10]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Explain the functions of a GIS. How GIS is different from other Database Management System (DBMS)? [5]
5. Describe the main ways of acquiring data for input into a GIS. [5]
6. Describe the different types of vector overlay operations with suitable examples. [5]
7. What are the different errors in GPS? How they can be corrected? [5]
8. Describe the basic elements of the raster data model. Why a local operation is also called a cell-by-cell operation? [5]

9.. Why do you most likely use a vector-based buffering operation, rather than a raster-based physical distance measure operation? Explain with examples. [5]

10. Explain what connectivity, adjacency and containment mean in vector topology. Provide an example for each. [5]
11. Describe spatial data infrastructure. Mention the contents of Metadata. [5]
12. Write short notes on: [2 x 2.5=5]
    a) Geoid and Ellipsoids
    b) Open GIS

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Bachelor Level/ Fourth Year/ Eighth Semester/Science — **Full Marks: 60**
Computer Science and Information Technology (CSC469) — **Pass Marks: 24**
(Decision Support System and Expert System) — **Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. Explain DSS Framework in detail. [10]
2. Discuss decision oriented diagnosis to DSS development. [10]
3. Why GDSS is important? Explain various group decision support situations. [10]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. How DSS differs from TPS? Explain. [5]
5. Discuss strategic impact grid. [5]
6. Explain various DSS project participants. [5]
7. Differentiate between DSS data and operating data. [5]
8. Explain architecture of expert systems in detail. [5]
9. Discuss fuzzy reasoning process. [5]
10. Discuss various types of fuzzy expert systems. [5]
11. Discuss expert system development life cycle. [5]
12. Write short notes on: [2 x 2.5=5]
    a. Cost-Effectiveness Analysis
    b. Executive DSS

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information (CSC470)	**Pass Marks: 24**
(Mobile Application Development)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. How do you define dimensions in mobile computing? Explain the architecture of mobile computing. [3+7]
2. Define mobile ecosystem. Describe the factors that affect the UI development with scenario. [2+8]
3. What are mobile agents? Explain the different mobile application testing approaches. [4+6]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Describe the WAP architecture for mobile development framework. [5]
5. Illustrate the significance of speech grammar in mobile UI development with example. [5]
6. Describe about transmission techniques. [5]
7. How do you handle synchronization and replication of mobile data? [5]
8. Why do we need location based mobile service? Explain. [5]

9.. How does mobile application handle active transactions? [5]

10. Describe about different user interface components. [5]
11. How do you distribute mobile application? Explain the process. [5]
12. Describe about different security threats and issues in mobile applications. [5]

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC471)	**Pass Marks: 24**
(Real Time Systems)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. Prove that "No on-line scheduling algorithm can achieve a competitive factor greater than 0.25 when the system is overloaded." [10]
2. Explain weighted Round-Robin service disciplines with required figure and example. [10]
3. Highlight on Firm Deadline model. What are its advantages and disadvantages? [8+2]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Differentiate between hard, soft and firm real time system. [5]
5. What are the conditions for a valid schedule? Define fixed, jitter and sporadic release times. [2+3]
6. Illustrate with an example how slack stealing can improve the average response time of aperiodic jobs. [5]
7. Explain memory locking in real time memory management. [5]
8. What is the fundamental difference between priority inheritance and priority ceiling protocol? What are the rules of basic priority-ceiling protocol? [1+4]
9. How can the analysis of sporadic and aperiodic interrupt system can be carried out? Explain. [5]
10. Illustrate on Medium Access-Control protocol.
11. Differentiate between fixed priority and dynamic priority scheduling. Briefly explain deferrable server. [2+3]
12. Write short notes on: [2 x 2.5=5]
    a. overlays
    b. Binary angular measure

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC472)	**Pass Marks: 24**
(Network and System Administration)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. How to administer Software-Defined Networking (SDN)? How do you perform disk partitioning for Linux installation over a desktop machine? [4+6]
2. What do you consider for Linux client and server installation? What are the benefits of open source database like MySQL? How do you perform job scheduling in Linux using cron/crontab/anacron? Explain with example. [4+3+3]
3. What is firewall? Explain the firewall tools used in Linux in network and application layer. How do you perform control and dataplane communication over SDN? [2+4+4]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Explain the DHCP lease renewable process. [5]
5. Discuss the DNS zone transfer process for dynamic updates. [5]
6. How do you implement proxy ACL for port level control of traffic in SQUID proxy? [5]
7. What is SMTP relay? Compare SMTP with IMAP. [1+4]
8. What is SPAM? How do you perform SPAM filtering at your mail server? [1+4]
9.. Describe about authenticating proxy server. [5]
10. Explain the procedure of web server virtual hosting. [5]
11. Describe the architecture of ONOS-SDN/IP. [5]
12. What is the difference between caching and replication? What are the major components of DHCP server configuration? [2+3]

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 60**
Computer Science and Information Technology (CSC473)	**Pass Marks: 24**
(Embedded Systems Programming)	**Time: 3 hours**

Section A

Attempt any two questions. (2 × 10 = 20)

1. Illustrate data processing instruction MOV with barrel shifter and ALU. Explain branch instructions B, BL, BX and BLX with examples and formats. [10]
2. Describe thumb instruction decompressor organization and thumb to ARM instruction mapping. [10]
3. Write C program to find squares of integers from 0 to 9 using a function (square) and convert square function by an assembly function to do the same actions. [10]

Section B

Attempt any eight questions. (8 × 5 = 40)

4. Illustrate major differences between RISC and CISC. Explain four major design rules for RISC philosophy implementation. [5]
5. Draw and explain ARM core dataflow model. [5]
6. Explain psr (program status register) byte fields. [5]
7. Explain thumb programmer's model. [5]
8. Discuss about basic C data types supported by ARM compliers. [5]

9.. Explain ATPCS (ARM – Thumb Procedure Call Standard) arguments passing during function call with a programming example. [5]

10. Explain 5 stage pipeline executing in ARM (ARM9TDMI). [5]
11. Explain firmware and bootloader. Show firmware execution flow. [5]
12. How components interact and function that determines the characteristics of a specific operating system. [5]

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Bachelor Level/ Fourth Year/ Eighth Semester/Science	**Full Marks: 80**
Computer Science and Information Technology (MGT474)	**Pass Marks: 32**
(International Business Management)	**Time: 3 hours**

Section A

Attempt any three questions. (3 × 10 = 30)

1. Define International Business. Explain the challenges of abroad business [10]
2. State the positive and negative impact of Globalization. [10]
3. Describe the Regional Economic Integration. Explain about economic. [10]
4. What is International Strategic management? Mention any two Modes of entering and operating in International markets. [10]

Section B

Attempt any ten questions. (10 × 5 = 50)

5. Write any five differences between Domestic and International Business. [5]
6. Explain about SAFTA. [5]
7. Describe about Political environment on international business. [5]
8. Write the meaning of intellectual property rights. [5]
9.. Explain about level of economic development. [5]
10. Mention the modes of payment in international trade. [5]
11. Explain the Factors affecting exchange rate. [5]
12. Explain in brief about franchising and management contract. [5]
13. Mention about the managing global supply chain. [5]
14. Describe the Geocentric approach in operation of International Business. [5]
15. Explain about global production strategies. [5]